

SANTHOSH T

Software Development Engineer | Python & Django Developer

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PROFESSIONAL SUMMARY

Results-driven Software Development Engineer and 2024 graduate (CGPA: 8.53) with hands-on experience in **Python** and **Django** web framework for building scalable, production-grade applications. Proficient in **Data Structures & Algorithms** and **Low-Level Design (LLD)** principles, with practical expertise in both **relational databases (MySQL)** and **non-relational databases (Vector DB, NoSQL)**. National-level Smart India Hackathon winner with a strong track record of delivering robust, tested, and optimized software solutions. Eager to contribute to Coverfox Insurance's engineering team with strong problem-solving, clean code practices, and a passion for continuous learning.

TECHNICAL SKILLS

Languages:	Python (Primary), Java, JavaScript, SQL
Web Frameworks:	Django, Django REST Framework, Spring Boot, React.js
Data Structures & Algo:	Arrays, Trees, Graphs, Dynamic Programming, Sorting, Searching, Time & Space Complexity
Low-Level Design:	OOP, SOLID Principles, Design Patterns (Factory, Singleton, Observer, Strategy), Class Diagrams
Relational DB:	MySQL — Schema Design, Normalization, Complex Joins, Indexing, Query Optimization
Non-Relational DB:	Vector DB (RAG/Embeddings), NoSQL Concepts, MongoDB (familiar)
Testing & DevOps:	Unit Testing, Debugging, Code Review, Git, Docker (Containerization), CI/CD Concepts
AI/ML Integration:	Groq LLM, Ollama, Spring AI, Retrieval-Augmented Generation (RAG)

PROJECTS

AI-Powered Post Calamity Management Portal

Full Stack | Python · Django · MySQL · Vector DB

React.js + Streamlit, Django, Vector DB, MySQL, Docker — Government-Grade Application

- Developed a **Django-based RESTful backend** for a full-stack disaster response portal managing relief registrations, compensation workflows, and real-time data across multiple agencies — applying LLD principles for modular, maintainable component design.
- Engineered a **Retrieval-Augmented Generation (RAG) chatbot** using Vector DB (non-relational) to parse unstructured SOS messages into targeted relief dispatches, implementing efficient data retrieval algorithms with optimized time complexity.
- Designed and managed **relational data models in MySQL** — normalized schemas with complex joins, indexing strategies, and query optimization to handle high-volume disaster data under load.
- Containerized the application using **Docker** and applied test-driven debugging to ensure reliability in production emergency-response environments.
- Applied **Low-Level Design patterns** (Factory, Strategy) for the alert-routing engine, ensuring scalability and clean separation of concerns across the codebase.

Personalised AI Learning Assistance Platform

Full Stack | Python · Django · MySQL · LLD

React.js, Django REST Framework, MySQL, Groq LLM, JWT Authentication

- Built **end-to-end full-stack application using Python and Django REST Framework** with AI-powered revision methodologies, personalized knowledge graphs, and adaptive content delivery for thousands of learning sessions.
- Applied **Data Structures (graphs for knowledge mapping, queues for scheduling)** and designed LLD class diagrams for the recommendation engine, ensuring efficient algorithm performance ($O(\log n)$ retrieval).
- Designed **relational MySQL schema** with user, course, progress, and assessment tables — implemented complex multi-table joins, foreign key relationships, and indexed queries for sub-100ms response times.

- **Integrated non-relational Vector DB** for semantic similarity search in AI-powered content recommendations, implementing embedding storage and nearest-neighbour retrieval.
- Implemented **SOLID design principles** across the Django application — dependency injection, interface segregation, and single-responsibility across 15+ Django models and views.

Smart India Hackathon – Automated Animal Identification

National Govt. Project | INCOIS

Python, AI/ML Models, Data Collection Pipelines, Algorithm Design

- **Led architecture design for an AI-based national government solution** — applied algorithmic thinking and problem-solving to design a scalable data pipeline handling large-scale image classification.
- **Implemented batch data processing pipelines in Python** — applied sorting, hashing, and search algorithms to preprocess and clean real-world datasets efficiently before model training.
- Won **national-level competition** demonstrating exceptional analytical thinking, problem decomposition, and delivery of a robust, deployable solution.

PROFESSIONAL EXPERIENCE

Product Support Engineer – Zoho Corporation

2024 – Present | Bangalore

- **Diagnosed and resolved complex application issues** in enterprise SaaS products — applied systematic debugging, log analysis, and root-cause analysis skills directly transferable to software optimization.
- **Collaborated with engineering teams** using structured code review and bug-reproduction workflows to resolve product issues, improving system stability and reducing ticket resolution time.
- Participated in **cross-functional problem-solving sessions**, translating customer-reported issues into clear technical specifications for engineering — sharpening analytical thinking and requirement analysis skills.

EDUCATION

B.E. – Electrical & Electronics Engineering

Batch: 2024

Bannari Amman Institute of Technology | CGPA: 8.53 / 10

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Computer Networks, Operating Systems
- 0 – 1 year experience range: Meets Coverfox's required experience criteria (0–1 years)

ACHIEVEMENTS

■ Smart India Hackathon 2022 – Winner	National Level Govt. Project, INCOIS
■ Healthware 2.0 – Winner	National Level Healthcare Tech Hackathon
■ CGPA: 8.53 / 10	First Class with Distinction — strong academic foundation
■ 2 National Hackathon Wins	Demonstrated problem-solving & algorithmic thinking under pressure

Open to full-time, on-site opportunities in Bangalore | Available immediately